



Toxic Epidermal Necrolysis After COVID-19 Vaccination: A Case Report and Review of Literature

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Following vaccination for COVID-19, various cutaneous adverse reactions (CARs) are reported. Here is an Asian male in late 50's who developed necrotic skin with mucosal involvement 10 days following booster dose of ChAdOx1 nCov-19 vaccination. Based on disease course and morphology, toxic epidermal necrolysis (TEN) was suspected. The patient developed respiratory distress and was intubated, intravenous immunoglobulin (IVIG) administered at 2 g/kg body weight following which skin lesions healed in fourth week, the patient was discharged after 50 days of intensive care unit (ICU) stay. Severe CARs are rare following vaccination, of two components in ChAdOx1nCoV-19 adenoviral vector vaccine, virotopes cause T-cell mediated granulysin and granzyme B release leading to epidermal detachment and mucosal involvement of conducting airways causing respiratory failure. CARs can also occur in whom first and second dose was uneventful. Supportive therapy and prevention of sepsis are mainstay of management. Though the use of IVIG has shown conflicting results, our case was successfully managed with IVIG.

Keywords: *adenoviral vector-based vaccine, adverse event, critically ill, cutaneous adverse reactions, SARS-CoV-2, toxic epidermal necrolysis*

Introduction

After emergency use authorization for different vaccines against severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection, various adverse events following immunization (AEFI) have been reported involving cardiac, neurological and hematological system. Similarly, skin is another organ where range of cutaneous adverse reactions (CARs) occur following SARS-CoV-2 vaccination. A systematic review and metanalysis showed pooled incidence of CARs as 5% (95%CI: 4%–6%, I² = 99%, $p < 0.001$) following SARS-CoV-2 vaccination.¹ They range from minor injection site reaction, delayed inflammatory reaction to tissue filler, flare of preex-

isting dermatoses, viral reactivation, to severe CARs (SCARs) like Steven Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), erythema multiforme and anaphylaxis.²

Here we report a case of TEN following a booster dose of adenoviral vector-based vaccine (ChAdOx1nCoV-19) vaccine.

Case Report

A 58-year-old male presented with pruritic lesion on forehead progressed to involve trunk, limbs with florid exfoliation. Only significant history was received booster dose ChAdOx1 nCov-19 vaccine 10 days prior to the onset of skin lesions. On Day-4 of

Received: June 6, 2023; Accepted: June 26, 2023.

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symptoms, he required admission to intensive care unit (ICU), due to difficulty in breathing and swallowing. Physical examination revealed coalesced skin lesions forming sheets of necrosed skin over face, torso and limbs with few areas showing bullae. Epidermal peeling involved 30%–40% body surface area with positive Nikolsky's sign. Mucosal involvement was present in the form of oral erosions, hemorrhagic crusting of lips, mucopurulent discharge from the eyes and glans (Fig. 1A and 1B). Investigations revealed mild transaminitis, aspartate aminotransferase 66 u/L (normal 5–40 u/L), alanine aminotransferase 101 u/L (normal 5–40 u/L), raised procalcitonin 1.15 ng/mL and normal total leucocyte count and renal function tests. Serology did not reveal any active infection, the severity of illness score for Toxic Epidermal Necrolysis (SCORTEN) score was 4 (58.3% mortality) on the day of admission, Naranjo algorithm score was 4, suggesting a possible association between vaccine and adverse drug reaction. On Day-10 patient intubated for respiratory distress and altered sensorium. Intravenous immunoglobulin (IVIG) initiated at 2 g/kg body weight on Day-13 following which clinical improvement noted 6 days later Day-19 in the form of non-progressive skin lesions. Supportive measures followed, skin and mucosal lesions healed at fourth week, weaned after mechanical ventilation for 28 days and discharged on Day-50. During follow-up, at the seventh day, the patient had healed skin lesion (Fig. 1C) and partial vision loss of left eye as sequelae to corneal epithelial defects at one-month.

Discussion

TEN is an extremely rare but life-threatening medical emergency. There are few case reports of TEN after measles-mumps-rubella, influenza, meningococcal-B and hantavirus vaccination.^{2,3} Following SARS-CoV-2 vaccination increasing number of CARs are described;^{4,6} however, only few cases of TEN are reported and most were following the first dose of vaccine (Table 1). Metanalysis has shown incidence of SCARs to be similar between the first and the second dose.^{3,6} Usually, recurrence of symptoms after booster dose occurs in those who experienced some reactions after the first dose, but CARs may also develop in those where the first two doses are uneventful as in our case.^{7,8} TEN usually occurs secondary to drugs or infection, a notable difference compared to vaccine-induced is early symptoms (< 2 weeks) whereas in drug exposure symptoms occur after 2–3 weeks.² Of the reported cases (Table 1), symptom onset was between 1–2 weeks which depends on several variables like type and dose of vaccine, immunological state, history of COVID-19 infection.^{9,10} Expression of virotopes on the surface of keratinocytes induces clonal expansion of CD8+ cytotoxic T lymphocytes resulting in Th-1 polarised response and release of granulysin and granzyme B causing keratinocyte apoptosis and detachment.^{11–13} There is no literature showing excipient causing severe delayed-type hypersensitivity reactions like TEN.¹³ Supportive therapy is mainstay of management; of reported cases (Table 1) most of them received corticosteroids, few



Fig. 1. (A) Showing oral mucosal ulcer with sloughing. (B) Necrosed skin over back of trunk with epidermal detachment. (C) Healed skin lesion over back of trunk. Fig. 1A and 1B shows clinical image taken on Day 4; and 1C shows clinical image taken at Day 7 post discharge from the hospital.

Table 1. Clinical and treatment details of patients who developed SJS or TEN following COVID-19 vaccination

NO	Age /Sex	Comorbidities	Type of vaccine and dose	SCARs	Clinical presentation and [day of onset after vaccination]	Treatment	Day of healing of skin lesion	Author/Reference
1	49/F	No	mRNA, 1 st	TEN	Fever, Skin eruptions [D7]	Etanercept	D-22	Bakir et al. ⁷
2	Middle age /F	No	mRNA, 2 nd	SJS	Fever, Red coloured bullae [D5]	Oral steroid	Not mentioned	Elboraey and Essa. ¹³
3	12/F	No	mRNA, 1 st	TEN	Fever, erythematous patches [D6]	IVIG	D-12	Siripattanamongkol et al. ³
4	46/F	Type 2 DM Hyperlipidaemia, Obesity	mRNA, 1 st	SJS	Oral discomfort, mucosal sloughing [D2]	Oral steroid	D-6	Padniewski et al. ²
5	86/F	Tinea Pedis on oral Terbinafine	mRNA, 3 rd	TEN	Painful pruritic rash on left arm [D1]	IV steroid	D-20	Han et al. ⁴
6	32/M	No	Inactivated virus vaccine, 2 nd	SJS	Blistering rash, erythematous plaque and punctate keratitis [D1]	Ocular steroid	D-10	Boualila et al. ¹¹
7	92/M ^{a,b}	Hypertension	Inactivated virus vaccine, 1 st	TEN	Painful skin lesions-necrosis [D20]	Supportive	Not mentioned	Seck et al. ⁶
8	76/M	No	Inactivated virus vaccine, 1 st	TEN	Skin lesions-blister [D1]	Oral steroid	D-14	Mardani et al. ⁵
9	49/F	Carcinoma breast	Inactivated virus vaccine, 2 nd	SJS	Ulcer in oral cavity and vagina [D3]	Oral steroid	D-14	Mansouri et al. ⁸
10	63/F	Diabetes mellitus on OHA, Mild plaque type psoriasis	Inactivated virus vaccine, 1 st	SJS	Bullous rash and lip ulcers [D1]	Oral steroids	D-7	Mansouri et al. ¹²
11	48/F	No	ChAdOx1 nCoV-19, 1 st	TEN	Erythematous eruption [D14]	Monoclonal antibody	D-28	Kherlopian et al. ⁹
13	65/M	No	ChAdOx1 nCoV-19, 2 nd	SJS	Mucocutaneous eruptions, DVT, Sagittal sinus thrombosis [D10]	Oral corticosteroid	Not mentioned	Aimo et al. ¹⁰
14	58/M ^c	Type 2 DM	ChAdOx1 nCoV-19, 3 rd	TEN	Pruritic lesion-blister and necrosis [D10]	IVIG	D-30	Our case

^a Organ failure; 92/M with renal insufficiency (creatinine clearance levels: 46.94 mL/min, and azotaemia: 1.10 g/L), electrolyte disorders (hyponatremia: 150 mmol/L), thrombocytopenia (77,000/mm³)^b Non-survivor^c Respiratory failure

BSA: body surface area; CARs: cutaneous adverse reactions; ChAdOx1 nCoV-19 vaccination: adenoviral vector-based vaccine; D: day of illness; DM: diabetes mellitus; DVT: deep venous thrombosis; ICU: intensive care unit; IVIG: intravenous immunoglobulin; mRNA: messenger ribonucleic acid; OHA: oral hypoglycaemic drugs; SARS-CoV-2: severe acute respiratory syndrome related virus; SCARs: severe cutaneous adverse reactions; TEN: toxic epidermal necrolysis; SJS: Steven Johnson syndrome.

only IVIG. Systemic therapy of TEN includes different combinations of corticosteroids, TNF- α inhibitor, cyclosporine, IVIG, and plasmapheresis.¹⁴ Short-time use of high-dose corticosteroids in early stages has been shown to reduce mortality without any negative impact on wound healing and infection risk. Benefits from IVIG is conflicting and some studies have shown mortality benefits with using a high dose.^{3,12} IVIG blocks the interaction of the Fas receptor with the Fas ligand a reported pathway for keratinocyte death. The use of Etanercept has a role if there is a pathogenic implication of TNF- α ; while, cyclosporine by its anti-apoptotic effects has shown to cause rapid cessation of disease progression and improved mortality in some cases.¹⁰ When disease is refractory to supportive and systemic steroid, plasmapheresis has been shown to be beneficial.^{10,12} SCARs are potentially life-threatening and represent contraindications for subsequent doses; it is not clear whether using different vaccine type for booster dose is advisable in such cases.⁵

Our case highlights successful management of extremely rare SCARs-TEN with extensive mucosal involvement and respiratory failure, following booster dose of ChAdOx1 nCoV-19 vaccination.

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